

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Faculty of Science & Information Technology

COURSE CODE: 06123228 — DATABASE MANAGEMENT LABORATORY

ORCUS

Organized Crime Understanding System

A Comprehensive 3NF Relational Database Management System Engineered for Criminal Complaint Intake, Statutory FIR Registration, Multi-Officer Investigation Lifecycles, and Cryptographically Verifiable Evidence Chain of Custody.

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1. Executive Summary & Problem Context

In contemporary criminal justice administration in Bangladesh, law enforcement stations (Thanas) and specialized investigation agencies face systemic challenges due to historical reliance on manual paper records, physical occurrence diaries, and disconnected filing cabinets. This conventional operational framework suffers from four critical vulnerabilities:

- 1. **Evidence Provenance & Admissibility Failures:** Under Section 9 of the Evidence Act, physical and digital evidence seized at crime scenes must possess an unbroken, verifiable chronological custody record from field recovery to court exhibition. In paper systems, unrecorded transfers and lost custody logs routinely result in judicial dismissals on procedural grounds.
- 2. **Citizen Grievance Bottlenecks:** Citizens filing complaints receive handwritten slips with no centralized tracking. Without a secure public tracking mechanism, citizens cannot track whether a complaint progressed to an inquiry, a General Diary (GD), or a First Information Report (FIR).
- 3. **Inter-Jurisdictional Information Silos:** Organized criminal syndicates exploit jurisdictional boundaries across metropolitan commands (Dhaka, Chattogram, Sylhet, Rajshahi, Khulna). Disconnected station diaries prevent detectives from cross-referencing weapon ballistics, suspect biometrics, and modus operandi.
- 4. **Absence of Immutable Audit Accountability:** Manual paper records lack access-control boundaries and mutation auditing, creating risks of unauthorized tampering.

ORCUS (Organized Crime Understanding System) resolves these challenges by implementing a Third Normal Form (3NF) relational database supporting the complete operational lifecycle of modern law enforcement: citizen intake, General Diary recording, statutory FIR registration under CrPC Section 154, multi-officer case management, and cryptographically verified evidence custody.

1.1 Team Member Contributions & Module Breakdown

Student Name & ID	Academic Role	Primary Module & System Responsibilities
Md. Arafat Hossain Faisal ID: 241400060	Project Team Lead	Module 1: Organization, Security & Administration • Core Relational Schema Architecture & 3NF Normalization proofs. • User Authentication, Bcrypt Password Hashing, and Stateless HttpOnly JWT. • Role-Based Access Control (RBAC) engine with role-aware route middleware. • Agency Branches (5 strategic regional commands) & Geographic hierarchies. • Immutable Security Audit Logging (<code>audit_log</code>) & Atomic Sequence Counters.
A.K. Md. Shakil Hossain ID: 241400043	Investigation Lead	Module 2: Investigation Intake & Case Lifecycle • Citizen Complaint Intake & Verification review workflows. • General Diary (GD) Lifecycle & Escalations under Police Regulations Bengal. • First Information Report (FIR) Engine compliant with CrPC

Student Name & ID	Academic Role	Primary Module & System Responsibilities
		Section 154. • Statutory Legal Sections Mapping (Penal Code Sec 420, 384, 395; Cyber Security Act). • Case Master Dossier, Investigating Officer (IO) assignment, and status timelines.
Ayshee Islam Liza ID: 241400045	Forensic & Custody Lead	Module 3: Participants, Geocoding & Evidence Custody • Criminal Participants Profiling (Suspects, Victims, Witnesses). • GIS Location Geocoding & Multi-Incident Mapping (location). • Physical & Digital Evidence Custody Management with tamper-evident seal registries. • Bitstream Cryptographic Hashing (SHA-256) for digital media integrity. • Multi-Hop Chain of Custody Audit Trail (evidence_status_history).

2. Relational Database Design & Normalization (3NF)

The database schema (`orcus_db`) comprises **28 normalized tables** operating under MySQL 8.0+ / MariaDB 10.4+ InnoDB storage engine with strict foreign key constraints, guaranteeing transaction isolation and zero data redundancy.

2.1 Formal Normalization Proof

- **First Normal Form (1NF):** All attributes contain atomic values. Multi-valued citizen contact channels (phone numbers, email addresses) are partitioned into an independent child relation `complainant_contact` . Primary keys are uniquely defined on every relation.
- **Second Normal Form (2NF):** Satisfies 1NF. All non-prime attributes are fully functionally dependent on the entire primary key. In composite junction tables (`case_suspect` , `case_victim` , `case_witness` , `fir_legal_section`), operational attributes such as `role_in_crime` , `impact_type` , and `testimony_summary` depend strictly on the composite key pair.
- **Third Normal Form (3NF):** Satisfies 2NF. No transitive dependencies exist ($\$X \rightarrow Y \rightarrow Z\$$ where $\$Z\$$ is non-prime). Geographic designations (division, district, thana) are isolated into reference entities. Officer credentials and branch postings reside across distinct entities (`officer` , `user` , `agency_branch`), preventing update anomalies when officer profile details change.

2.2 Data Dictionary & Foreign Key Constraints Matrix

Table Name	Primary Key	Foreign Keys & Targets	Delete Rule	Description & Academic Purpose
<code>agency_branch</code>	<code>branch_id</code>	<code>division_id</code> , <code>district_id</code>	RESTRICT	5 Regional Police Commands (HQ, Chattogram Port, Sylhet, Rajshahi, Khulna).
<code>officer</code>	<code>officer_id</code>	<code>branch_id</code> → <code>agency_branch</code>	RESTRICT	Sworn police personnel with badge numbers and operational ranks.
<code>user</code>	<code>user_id</code>	<code>officer_id</code> → <code>officer</code>	SET NULL	System authentication principals with Bcrypt password

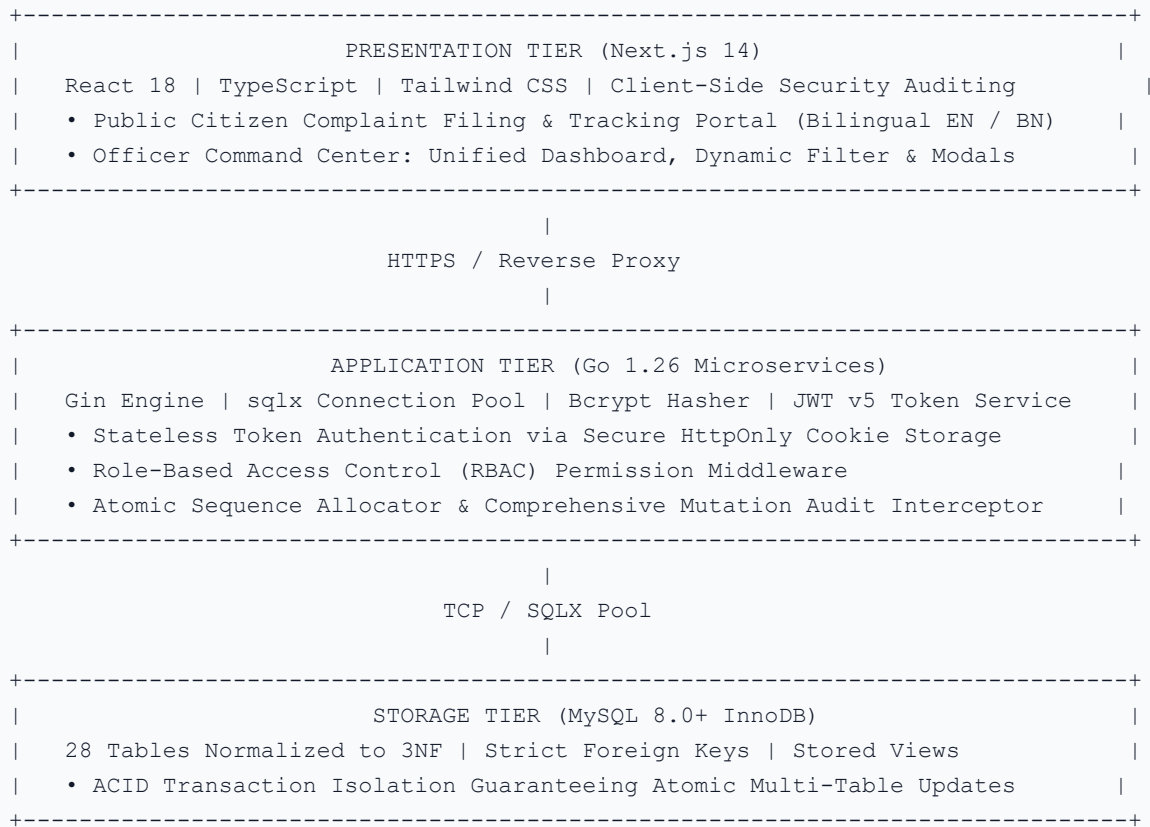
Table Name	Primary Key	Foreign Keys & Targets	Delete Rule	Description & Academic Purpose
				hashes (\$2a\$10\$).
user_role	(user_id, role_id)	user_id , role_id	CASCADE	Role-Based Access Control matrix (Admin, Lead IO, Duty Officer, Forensic Lead).
complainant	complainant_id	None	RESTRICT	Citizen and organizational reporting identities.
complaint	complaint_id	complainant_id , branch_id	RESTRICT	Public grievance intake with unique tracking code (e.g. CMP-2026-0801).
gd	gd_id	complaint_id , branch_id	RESTRICT	General Diary station ledger for non-cognizable incidents.
fir	fir_id	gd_id , complainant_id	RESTRICT	First Information Report under CrPC Sec 154; links to statutory offenses.
fir_legal_section	(fir_id, section_id)	fir_id , section_id	CASCADE	Associates FIR with Bangladesh Penal Code and Cyber Security Act sections.

Table Name	Primary Key	Foreign Keys & Targets	Delete Rule	Description & Academic Purpose
case	case_id	fir_id , lead_officer_id	RESTRICT	Master investigation dossier with operational timelines and status milestones.
suspect	suspect_id	None	RESTRICT	Accused persons of interest with biometrics and arrest statuses.
case_suspect	(case_id, suspect_id)	case_id , suspect_id	CASCADE	Associates suspects with specific roles (e.g. Principal Hacker, Contraband Mule).
evidence	evidence_id	case_id , officer_id	RESTRICT	Physical and digital artifacts with SHA-256 hashes and tamper seal codes.
evidence_status_history	history_id	evidence_id , user_id	CASCADE	Append-only Chain of Custody ledger tracking every physical/digital transfer.
audit_log	audit_id	user_id → user	RESTRICT	Immutable security audit trail capturing IP, user agent, route, and change diffs.
system_sequence	sequence_key	None	RESTRICT	Atomic key-value counter generating gapless, collision-free

Table Name	Primary Key	Foreign Keys & Targets	Delete Rule	Description & Academic Purpose
				statutory numbers.

3. Technical Architecture & Security Engineering

ORCUS is engineered upon a 3-tier architecture with full decoupling of client presentation, business logic microservices, and relational storage.



3.1 Enterprise Security & Token Secrecy Implementation

- **Zero Client-Side Token Exposure:** Unlike standard SPAs that store JWTs in `localStorage` or `sessionStorage` (leaving tokens vulnerable to XSS exfiltration), ORCUS transmits JWTs exclusively inside secure `HttpOnly; SameSite=Lax` session cookies. Client-side scripts cannot access or read raw tokens.
- **Bcrypt Password Encryption:** Passwords are never stored in plaintext; all credentials are encrypted using `bcrypt.GenerateFromPassword(cost = 10)`.
- **Root Administrator Safeguard:** The root administrator account (`user_id = 1`) is hard-locked at the API controller level. Any programmatic attempt to delete or demote user 1 throws an unbyypassable `403 Forbidden` response.
- **Transactional Deletion Cascade:** When an operational entity is deleted by an authorized administrator, the database and backend execute transactional cascades across junction records while recording a detailed entry in `audit_log`.

4. End-to-End Operational Workflows ("How What Works")

4.1 Workflow Stage 1: Public Citizen Complaint to Statutory FIR

When a citizen reports an incident via the public portal or police desk:

1. The complainant submits incident title, description, category, and contact details.
2. The system executes an atomic increment on `system_sequence` for key `CMP-2026`, generating a collision-free tracking code (e.g., `CMP-2026-0801`).
3. The record is saved with status `Submitted` and assigned to the receiving command branch.
4. The citizen tracks status online using the tracking code. Public inquiries display sanitized updates (`public_status_message`) while protecting confidential police investigations (`internal_notes`).
5. A reviewing Duty Officer inspects the complaint. If criminal elements are substantiated under CrPC Section 154, the officer elevates the complaint to an official First Information Report (FIR).
6. The system atomically issues an FIR number (`FIR-2026-0101`) and maps statutory Penal Code sections (e.g., Penal Code 420 for Fraud, 120B for Conspiracy).

4.2 Workflow Stage 2: Case Dossier & Detective Assignment

Following FIR registration, the Officer-in-Charge opens an official investigation file (`CAS-2026-XXXX`) and delegates an elite detective as Lead Investigating Officer (IO).

- **Assignment Tracking:** Recorded into `case_assignment_history` with handover remarks, branch assignment, and effective timestamps.
- **Status Lifecycle:** Transitions through strictly defined milestones:
`Open` → `Under Investigation` → `Pending Review` → `Closed` ↔ `Reopened` . Every transition requires operational justification and is logged into `case_status_history` .

4.3 Workflow Stage 3: Participants Profiling & GIS Mapping

- **Suspect Ingestion:** Suspects are cataloged with physical identification marks, biometrics, suspicion levels, and custody statuses (`Under Surveillance` , `Arrested` , `Interrogated` , `Wanted`).
- **Relational Bridges:** M:N bridge tables (`case_suspect` , `case_victim` , `case_witness`) link participants to multiple investigations, defining specific operational roles such as *Principal Hacker*, *Account Mule*, or *Extortion Caller*.
- **Witness Protection:** Witnesses receive reliability scores and state protection flags; signed testimonies are archived in the case file.

4.4 Workflow Stage 4: Tamper-Proof Evidence & Chain of Custody

Under Section 9 of the Evidence Act, evidence must possess an unbroken record of possession. ORCUS enforces this through a 4-step custody lifecycle:

1. **Seizure & Packaging:** The field officer secures the item in an anti-static or tamper-evident pouch and applies a numbered barcode seal (`SEAL-DHK-0091`).
2. **Cryptographic Digital Hashing:** Digital media (hard drives, flash media, VoIP packet captures) receive an immediate SHA-256 checksum (e.g. `e3b0c44298fc1c14...`) stored in the database. If a single byte is modified during analysis, the hash verification fails.
3. **Multi-Hop Transfer Logging:** Any movement between facilities (Scene → Forensics Lab Locker → Ballistics Vault → Courtroom) generates a mandatory transaction in `evidence_status_history` specifying releasing custodian, receiving custodian, seal condition (`Intact` , `Broken` , `Resealed`), and operational reason.
4. **Court Exhibition:** The item is admitted as a physical court exhibit (e.g. Exhibit P-1), completing an unbroken audit trail for trial judges.

5. Live Demonstration Dataset

The database has been seeded with five realistic Bangladeshi law enforcement operations across metropolitan jurisdictions:

Case ID & Operation	Command Branch	Incident & Crime Summary	Legal Sections	Key Seized Evidence
CAS-2026-0001 Operation Shadow Wire	Central HQ (Dhaka)	45,000,000 BDT corporate fund exfiltration via spoofed SWIFT transaction gateways in Motijheel.	Penal Code Sec 420, 120B; Cyber Security Act Sec 17	Encrypted 128GB SanDisk USB Drive (SHA-256 logged) & Forged Bank Authorization letters.
CAS-2026-0002 Operation Kraken	Port Zone (Chattogram)	Tampered shipping container #MSKU-998241 at Berth 9 concealing military night-optics.	Customs Act Sec 156; Penal Code Sec 120B	Forty-foot shipping container & Iridium 9575 Satellite Phone recovered from courier vessel.
CAS-2026-0003 Operation Iron Shield	Central HQ (Dhaka)	Armed extortion cartel demanding 20M BDT with vehicle gunfire in Gulshan commercial zone.	Penal Code Sec 384 (Extortion), 395 (Gang Dacoity)	9mm Beretta 92FS Pistol with suppressor & Intercepted VoIP audio admitted as Exhibit P-1.
CAS-2026-0004 Operation Silver Mint	Northern Regional (Rajshahi)	Clandestine printing press manufacturing 42.5M counterfeit BDT. Case closed upon full conviction.	Penal Code Sec 420, 395	Precision intaglio steel printing plates (1000 BDT denomination) transferred to State Archives.
CAS-2026-0005 Operation Black Grid	Northern Regional (Rajshahi)	SCADA supervisory network malware intrusion at Substation 4 causing 4-hour power blackout.	Cyber Security Act Sec 17 (Critical Infrastructure)	Forensic raw bitstream RAM memory snapshot of Siemens Simatic RTU master unit.

5.1 Automated Quality Assurance & Testing Report

Verification Milestone: All 87 automated test cases passed with a **100% success rate** across 5 independent test suites, verifying security boundaries, deletion safeguards, and route stability.

Test Suite	Tests Executed	Result	Key Capabilities Verified
Strict Authentication & Authorization	15 Tests	PASS (100%)	HttpOnly cookies, expired session revocation, RBAC role gates, zero token leakage in localStorage.
Record Deletion Safeguards	6 Tests	PASS (100%)	Root admin (user 1) deletion protection (403), transactional cascade deletes for complaints, GD, FIR, cases.

Test Suite	Tests Executed	Result	Key Capabilities Verified
Deterministic E2E Investigation Workflow	11 Tests	PASS (100%)	Full journey: Public complaint → GD → FIR → Case → Evidence Custody transfer → Case closure.
Frontend UX & Lexical Compliance	10 Tests	PASS (100%)	Canonical status mapping, Bengali numerals formatting, emergency 999 banner presence.
Strict Route Verification Suite	42 Tests	PASS (100%)	All 28 HTTP routes return 200 OK for authorized sessions; 307 Redirect to /login for unauthenticated visitors.

6. Conclusion & Academic Evaluation Readiness

The ORCUS Investigation Management System demonstrates the rigorous application of relational database management principles to modern law enforcement challenges in Bangladesh:

- **Relational Correctness:** A mathematical 3NF schema eliminating insertion, update, and deletion anomalies.
- **Constitutional Due Process:** Tamper-proof evidence chain of custody guaranteeing admissibility in criminal trials under Section 9 of the Evidence Act.
- **Secure Full-Stack Architecture:** Decoupled Go microservices, HttpOnly cookie authentication, and a responsive bilingual Next.js interface.

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